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Aloe-emodin: A review of its pharmacology, toxicity, and pharmacokinetics

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Aloe-emodin is a naturally anthraquinone derivative and an active ingredient of Chinese herbs, such as *Cassia occidentalis*, *Rheum palmatum* L., *Aloe vera*, and *Polygonum multiflorum* Thunb. Emerging evidence suggests that aloe-emodin exhibits many pharmacological effects, including anticancer, antiviral, anti-inflammatory, antibacterial, antiparasitic, neuroprotective, and hepatoprotective activities. These pharmacological properties lay the foundation for the treatment of various diseases, including influenza virus, inflammation, sepsis, Alzheimer's disease, glaucoma, malaria, liver fibrosis, psoriasis, Type 2 diabetes, growth disorders, and several types of cancers. However, an increasing number of published studies have reported adverse effects of aloe-emodin. The primary toxicity among these reports is hepatotoxicity and nephrotoxicity, which are of wide concern worldwide. Pharmacokinetic studies have demonstrated that aloe-emodin has a poor intestinal absorption, short elimination half-life, and low bioavailability. This review aims to provide a comprehensive summary of the pharmacology, toxicity, and pharmacokinetics of aloe-emodin reported to date with an emphasis on its biological properties and mechanisms of action.

KEYWORDS

aloe-emodin, mechanisms, pharmacokinetics, pharmacology, toxicology

1 | INTRODUCTION

Aloe-emodin (1,8-dihydroxy-3-hydroxymethyl-anthraquinone; Figure 1) is a natural anthraquinone derivative that occurs in many widely

used Chinese medicinal herbs, such as *Cassia occidentalis*, *Rheum palmatum* L., *Aloe vera*, and *Polygonum multiflorum* Thunb, that are used as traditional medicines in many countries (Panigrahi, Mudiam, Vashishtha, Raisuddin, & Das, 2015; Huang, Lu, Shen, Chuang, & Ong, 2007; Hamman, 2008; Yi, Leung, Lu, Zhang, & Chan, 2007). Currently, aloe-emodin has attracted much attention for its remarkable antineoplastic activity on different tumor cells, such as lung, gastric, hepatic, melanoma, breast, and colon cancer cells (Huang et al., 2013; Lee, Lin, Yang, Leung, & Chang, 2006; Li et al., 2016; Lu, Shen, Chung, & Ong, 2007; Lu, Shen, Ong, & Chung, 2007; Suboj, Babykutty, Srinivas, & Gopala, 2012; Tabolacci et al., 2015). Furthermore, aloe-emodin has been demonstrated to exhibit a wide spectrum of pharmacological effects, such as antiviral, anti-inflammatory, antibacterial, antiparasitic, neuroprotective, and hepatoprotective (Divya, Panonnummal, Gupta, Jayakumar, & Sabitha, 2016; Hu, Zhang, Meng,

Abbreviations: Ach, acetylcholine; AChE, acetylcholinesterase; Akt, protein kinase B; ALP, alkaline phosphatase; CAPN, calpain; CARP, caspase-8 and -10-associated RING protein; ER, endoplasmic reticulum; ERK1/2, extracellular signal-regulated kinases 1 and 2; FAK, focal adhesion kinase; GPx, glutathione peroxidase; HER, human epidermal growth factor receptor; HSP90, heat shock protein 90; HUVEC, human umbilical vein endothelial cells; IFN, interferon; IL, interleukin; ILK, intracellular integrin-linked kinase; JNK, Jun-N-terminal kinase; MAPK, mitogen-activated protein kinase; MDA, malondialdehyde; MMP, matrix metalloproteinase; mTORC, mammalian target of rapamycin complex; nano-AE, aloe-emodin-encapsulated nanoliposomes; NAT, N-acetyltransferase; NF-κB, nuclear factor-κB; NMDA, N-methyl D-aspartate; NPC, nasopharyngeal carcinoma; PCNA, proliferating cell nuclear antigen; PDT, photodynamic therapy; PKC, protein kinase C; PRDX, peroxiredoxins; RGCs, retina ganglion cells; ROS, reactive oxygen species; SOD, superoxide dismutase; TG 2, transglutaminase 2; TNF-α, tumor necrosis factor-α; UBE3A, ubiquitin-protein ligase E3A; VEGF, vascular endothelial growth factor

Wang, & Wang, 2014; Li et al., 2014; Tao et al., 2014; Wang, Chung, Ho, Wu, & Chang, 1998; Woo et al., 2002). Hence, it has been widely used in the treatment of influenza virus, inflammation, sepsis, Alzheimer's disease, glaucoma, malaria, liver fibrosis, psoriasis, Type 2 diabetes, growth disorders, and several types of cancers.

However, aloe-emodin could also lead to phototoxicity, hepatotoxicity, and kidney toxicity, particularly in high doses and with long-term use. This review discusses the current knowledge of the pharmacology, toxicity, and pharmacokinetics of aloe-emodin in the past few decades to reveal their therapeutic potentials and gaps offering opportunities for future research.

2 | PHARMACOLOGY

2.1 | Anticancer activity

2.1.1 | Lung cancer

Lee, Hsu, Liu, and Wu (2001) suggested that aloe-emodin could inhibit the proliferation of CH27 cells through the Bax and Fas death pathway. Furthermore, aloe-emodin could induce DNA damage and chromatin condensation through generation of reactive oxygen species (ROS) in human lung nonsmall carcinoma H460 cells. The molecular mechanisms were associated with caspase-3-dependent pathway but not apoptosis-inducing factor apoptotic pathway (Lai, Hour, Leung, Yang, & Lee, 2007; Lee, Wu, & Chang, 2005; Yeh, Wu, & Lee, 2003). In another experiment, aloe-emodin was shown to induce apoptosis and changes in cell morphology, which were in part regulated through its effect on protein kinase C (PKC) δ -mediated cytoskeleton in H460 cells (Chang et al., 2012). Wu, Zhang, Gao, and Li (2017) demonstrated that aloe-emodin nanoparticles significantly suppressed the proliferation of squamous cell carcinoma (SCC) cell line and induced cell cycle arrest and apoptosis via mitogen-activated protein kinase (MAPKs) and phosphoinositide-3 kinase (PI3K)/protein kinase B (AKT) pathways, indicating aloe-emodin nanoparticles might be a novel therapeutic strategy for management of human lung squamous carcinoma.

2.1.2 | Gastric cancer

Chen, Lin, Chang, Fang, and Lin (2007) for the first time demonstrated that aloe-emodin exhibited the anticancer effect against AGS and NCI-N87 gastric carcinoma cells in a dose- and time-dependent manner by activating the expression of caspase-3 and suppressing the activity of the casein kinase II. Guo et al. (2007) show that aloe-emodin inhibited the growth of MGC-803 cells in a dose-dependent manner with an increase in S phase. The alkaline phosphatase (ALP) activity, an indicator of cell differentiation, was found to decrease significantly. However, SGC-7901 cells were arrested at G2/M phase in a dose-dependent manner with the decreased of S phase through PKC pathway and c-Myc suppression (Guo, Xiao, Liu, Gong, & Le, 2008). Furthermore, the proliferation of MKN45 cells was significantly inhibited

after aloe-emodin treatment by leading to cell-cycle arrest at G0/G1 phase (Chihara et al., 2015).

Additionally, aloe-emodin was used as a photosensitizer to explore the effect of photodynamic therapy (PDT) in human gastric cancer SGC-7901 cells. Li et al. (2016) explored the combined effect of gene transfection with PDT on SGC-7901 cells using aloe-emodin-encapsulated nanoliposomes. The results revealed that aloe-emodin-encapsulated nanoliposomes-mediated r-caspase-3 gene transfection coupled with PDT remarkably inhibited the proliferation of SGC-7901 cells. The main mechanisms were attributed to aloe-emodin-induced apoptosis and necrosis of gastric cancer cells (Li et al., 2016). Similarly, Lin et al. (2017) also showed that aloe-emodin-induced PDT exerted an inhibitory effect on SGC-7901 cells in dose-dependent and energy-dependent manners. Meanwhile, an increase in caspase-9 and caspase-3 expression was also observed, which indicated that the mitochondrial pathway for apoptosis may be involved (Lin et al., 2017).

2.1.3 | Liver cancer

Kuo, Lin, and Lin (2002) was the first to investigate the anticancer effect of aloe-emodin in two human liver cancer cell lines (HepG2 and Hep3B). The results indicated that aloe-emodin induced p53 expression and was accompanied by induction of p21 expression that was associated with a cell-cycle arrest in G1 phase in HepG2 cells. In addition, aloe-emodin markedly increased the expression of Fas/apoptosis antigen 1 (APO1) receptor and Bax. In contrast, aloe-emodin inhibited the proliferation of Hep3B cells through a p21-dependent manner that did not cause cell-cycle arrest or increase the level of Fas/APO1 receptor but rather promoted apoptosis by enhancing expression of Bax (Kuo et al., 2002). Lu, Shen, Chung et al. (2007) found that aloe-emodin was more cytotoxic and effective in inducing apoptosis and cell cycle arrest in HepG2 cell line than its analog emodin. Treatment with aloe-emodin also enhanced the intracellular level of ROS and oxidation of peroxiredoxin-2 and peroxiredoxin-4 (Lu, Shen, Chung, & Ong, 2007). Furthermore, aloe-emodin-induced apoptotic cell death of HepG2 cells was mediated via marked oxidative stress and sustained Jun-N-terminal kinase (JNK) activation (Lu, Shen, Ong, & Chung, 2007). Lin et al. (2011) indicated that aloe-emodin induced p53-independent apoptosis in Hep3B cells through the caspase-8-mediated activation of the mitochondrial death pathway by decreasing the stability of caspase-8 and -10-associated RING protein mRNAs. It has been shown that aloe-emodin exhibited potent antiproliferative effects on Huh-7 cells via downregulation of calpain-2 and ubiquitin-protein ligase E3A proteins (Jeon, Jeon, & Nam, 2012).

2.1.4 | Breast cancer

Breast cancer has one of the highest mortality rates of cancer among women worldwide (Hartmann, Reimer, & Gerber, 2011). Estrogen plays an important role in early breast cancer development (Walker, 1999). Previous studies showed that aloe-emodin exerted estrogenic

activity and modulated breast cancer cell proliferation as phytoestrogen compound (Kang et al., 2008; Matsuda, Shimoda, Morikawa, & Yoshikawa, 2001). Chen et al. (2016) demonstrated that aloe-emodin-PDT could inhibit cell proliferation, adhesion, migration, invasion, and induce apoptosis in breast cancer MCF-7 cells. The mechanisms of apoptosis were involved in mitochondrial and endoplasmic reticulum (ER) death pathways (Chen et al., 2016). Huang et al. (2013) indicated that aloe-emodin could suppress the proliferation of human breast cancer cell lines (MCF-7 and MDA-MB-453) by downregulating ER α protein levels, thereby suppressing ER α transcriptional activation. Moreover, aloe-emodin treatment led to the dissociation of heat shock protein 90 and ER α and increased ER α ubiquitination (Huang et al., 2013). Similarly, cotreatment of aloe-emodin with tamoxifen suppressed the expression of ER α , Ras, and extracellular signal-regulated kinases 1 and 2 (ERK1/2) and the phosphorylation of ERK1/2, PI3K, and mammalian target of rapamycin (mTOR) by blocking Ras/ERK and PI3K/mTOR pathways in MCF-7 cells (Tseng et al., 2017). Human epidermal growth factor receptor (HER)-2-overexpressing breast cancer has a high risk of treatment failure because it spreads rapidly and is highly metastatic as well as drug resistant (Adamczyk et al., 2015). It was shown by Ma et al. (2016) for the first time that aloe-emodin inhibited the expression of HER-2 through the downregulation of YB-1 via the downregulation of the intracellular integrin-linked kinase/Akt/mTOR signaling pathway in HER-2-overexpressing breast cancer cells, indicating that aloe-emodin could be used for clinically treating patients with HER-2 overexpression in the future.

2.1.5 | Melanoma cancer

Aloe-emodin treatment dose dependently suppressed the growth of human malignant melanoma A375.S2 cells by inhibiting the mRNA expression and enzyme activity of N-acetyltransferase (NAT; Lin et al., 2005). Radovic et al. (2012) demonstrated that aloe-emodin reduced the growth of mouse B16 and human A375 melanoma cells by two different signaling pathways. The cytostatic effect in B16 cells could be attributed to differentiation-inducing property of aloe-emodin through ROS production via activation of ERK1/2. However, aloe-emodin exposure dramatically inhibited ERK in A375 cells could be the cause of the detected apoptosis (Radovic et al., 2012). Aloe-emodin also exhibited an inhibitory effect on B16-F10 melanoma cells proliferation and metastatic potential in a time-dependent manner through the induction of cell differentiation. The possible mechanisms of action were involved in the activation of the transamidating form of TG2 and the reduction of matrix metalloproteinase (MMP)-9 activity (Tabolacci et al., 2010). Furthermore, aloe-emodin possessed remarkable antineoplastic and immunomodulatory properties against human melanoma cells (A375, M14, SK-MEL-110, and SK-MEL-28 cells) by increasing the expression of inflammation-associated factors such as interleukin (IL)-2, IL-12, GM-CSF, and interferon (IFN)- γ , indicating that this drug can be used as a promising agent for differentiation therapy of cancer (Tabolacci et al., 2015).

2.1.6 | Leukemia

Leukemia is one of the most deadly cancers in the world. Chemoprevention is a promising way to prevent the development of such cancers. It was shown for the first time that aloe-emodin inhibited the proliferation of human leukemia HL-60 cells and induced G2/M arrest, and apoptosis underwent activation of caspase-3 (Chen, Hsieh, Chang, & Chung, 2004). Similarly, aloe-emodin could suppress U937 cell proliferation in a time- and dose-dependent manner, enhance MMP secretion, and induce adhesion and migration abilities, indicating that it could be a potential candidate as a differentiation-inducing agent for the therapeutic treatment of leukemia (Tabolacci et al., 2011).

2.1.7 | Colon cancer

Nuclear factor- κ B (NF- κ B) is an important transcription factor for controlling MMP-2/9 and vascular endothelial growth factor (VEGF) gene expression. Suboj, Babykutty, Srinivas et al. (2012) reported that aloe-emodin effectively inhibited the migration and angiogenesis of WiDr colon adenocarcinoma cells through downregulating the mRNA and protein levels of MMP-2/9, RhoB, and VEGF. All these effects could be due to the inhibition of nuclear translocation and DNA binding of NF- κ B (Suboj et al., 2012). Furthermore, aloe-emodin inhibited the proliferation of the colon cancer cell lines (SW 480, SW 620, and RKO) in a concentration-dependent manner by arresting the cell cycle at the G2/M phase and inhibiting the expression of p21 and cyclin B1 proteins. Inhibition of the cell cycle and proliferation was carried out via downregulating the phosphorylation of ERK1/2 and activating stress-activated protein kinases/JNK and p38MAPK (Suboj, Babykutty, Srinivas, & Gopala, 2012). Shimp et al. (2014) investigated the cancer chemopreventive effects of aloe-emodin in an Apc-deficient multiple intestinal neoplasia mouse model. Aloe-emodin significantly reduced the number of colorectal tumors and decreased the mean MIB-5-labeling index. These results suggested that aloe-emodin may have chemopreventive effects against development of colorectal tumors in multiple intestinal neoplasia mice, possibly in part by reducing cell proliferation in colorectal mucosa (Shimp et al., 2014).

2.1.8 | Cervical cancer

Cellular ALP is increasingly recognized as important markers for monitoring tumor cell behavior in human malignancies. The measurement of ALP activity is usually used to determinate the effect of inducing the differentiation of anticancer reagents (Siddiqui & Kim, 1984; Xiao et al., 2006). Compared with treatment with aloe-emodin or radiation alone, the combination of aloe-emodin and radiation suppressed the proliferation of HeLa cells in a concentration- and time-dependent manner, induced G2/M and S phase cell cycle arrest, and improved ALP activity (Luo et al., 2014). Similarly, Guo et al. (2007) demonstrated that aloe-emodin (2.5–40 μ M) could inhibit the growth and induce G2/M phase arrest of HeLa cells in a dose-dependent manner. More importantly, the ALP activity was found to be increased and

accompanied by the inhibition of proliferating cell nuclear antigen expression. In addition, aloe-emodin suppressed the expression of PKC α and c-Myc, suggesting that this effect may be through the PKC α pathway (Guo, Xiao, Zhang, et al., 2007).

2.1.9 | Tongue cancer

Treatment with aloe-emodin (30 μ M) significantly induced DNA damage in SCC-4 human tongue cancer cells followed by the inhibition of DNA repair-associated gene expressions including ataxia telangiectasia-mutated gene, ataxia telangiectasia-mutated and Rad3-related, breast cancer gene 1, DNA-dependent protein kinase, and O⁶-methylguanine-DNA-methyltransferase (Chen et al., 2010). In another experiment, aloe-emodin (30 μ M) was shown to inhibit the proliferation of SCC-4 cells through S-phase arrest and apoptosis in a dose- and time-dependent manner. Furthermore, aloe-emodin induced ROS and Ca²⁺ production, ER stress, dysfunction of mitochondria, caspase-9 and caspase-3 activation through the Fas/death receptor, mitochondria, and caspase cascade (Chiu et al., 2009).

2.1.10 | Glioma

Aloe-emodin (40 μ M) inhibited the proliferation and S-phase progression in SVG and U-373MG cells. It also suppressed the activity of PKC and reduced the protein content of the PKC isozymes by regulating the PKC signaling pathway (Acevedo, Russell, & Patel, 2004). Mijatovic et al. (2005) showed that aloe-emodin could induce apoptosis and autophagy in the C6 glioma cell line. However, aloe-emodin did not affect the activation of MAPK p38 and JNK but markedly inhibited the activation of ERK1/2 (Mijatovic et al., 2005). Aloe-emodin has been reported to decrease the cell viability of U87 malignant glioma cells in a time- and dose-dependent manner, causing cell cycle arrest at S phase and promotes the loss of mitochondrial membrane potential. These data demonstrated that aloe-emodin could be a chemotherapeutic drug for the treatment of human brain cancer (Ismail et al., 2013).

2.1.11 | Other cancer

It was shown by He, Yan, Mo, and Liang (2008) for the first time that aloe-emodin significantly inhibited invasion, migration, and adhesion of human ovarian cancer cell line HO-8910PM. It also reduced the protein and mRNA expression of focal adhesion kinase (FAK). The possible mechanistic explanation for the inhibitory effect on metastasis potential is involved in FAK expression (He et al., 2008). Moreover, aloe-emodin inhibited cell viability and induced G2/M arrest and apoptosis in T24 human bladder cancer cells through the activation of p53, p21, Fas/APO-1, Bax, and caspase-3 (Lin et al., 2006). A similar study showed that aloe-emodin induced human nasopharyngeal carcinoma cells in G2/M phase by increasing levels of Cyclin B1 bound to Cdc2 and caused an increase in apoptosis of nasopharyngeal carcinoma cells via caspase-8-mediated activation of the mitochondrial death pathway (Lin et al., 2010). Liu et al. (2012) reported that aloe-emodin effectively

suppressed the proliferation and anchorage-independent growth of the androgen refractory prostate cancer cell line PC3 through its targeting of mTORC2. In another experiment, aloe-emodin was shown to inhibit the proliferation of esophageal cancer cell TE1 and reduces the number of TE1 cells in S phase through the inhibition of AKT, ERK, and their downstream molecules, thus regulating cyclin D1 transcription (Chang et al., 2016). Moreover, aloe-emodin-mediated PDT effectively decreased the viability of human osteocarcinoma MG-63 cells and induced autophagy and apoptosis through activation of the ROS-JNK signaling pathway (Tu et al., 2016).

2.2 | Antivirus activity

Influenza A belongs to Orthomyxoviridae virus family, exhibiting 16 hemagglutinin subtypes and 9 neuraminidase subtypes. Li et al. (2014) demonstrated that aloe-emodin could reduce virus-induced cytopathic effect and inhibit replication of influenza A in a concentration-dependent manner in MDCK cells. Proteomics and western blot of MDCK cells indicated aloe-emodin upregulated galectin-3 and thioredoxin, as well as downregulated nucleoside diphosphate kinase A (Li et al., 2014). Lin et al. (2008) investigated the antiviral activity of aloe-emodin against Japanese encephalitis virus and enterovirus 71 using plaque reduction assays in human HL-CZ promonocyte cells and TE-671 medulloblastoma cells. The results showed that aloe-emodin exhibited dose- and time-dependent inhibitory activities on the replication of Japanese encephalitis virus and enterovirus 71 in TE-671 and HL-CZ cells via IFN signaling responses. Furthermore, aloe-emodin significantly induced the expression of IFN- α , protein kinase R, and 2'-5'-oligoadenylate synthetase, the activation of interferon stimulating responsive element and IFN- γ -activation site promoters, and the production of NO (Lin et al., 2008).

2.3 | Anti-inflammatory Activity

Aloe-emodin markedly suppressed the production of NO, IL-6, and IL-1 β in lipopolysaccharide-stimulated RAW 264.7 cells. Moreover, it inhibited inducible nitric oxide synthase protein expression, I κ B α degradation, and the phosphorylation of ERK, p38, JNK, and Akt. The results demonstrated that aloe-emodin exerted an anti-inflammatory effect through a likely mechanism involving a decrease in proinflammatory cytokine production in lipopolysaccharide-induced RAW 264.7 macrophages via inhibition of NF- κ B, MAPK, and PI3K pathways (Hu et al., 2014). In acute liver injury rats model induced by carbon tetrachloride, aloe-emodin had some protective effect not only against hepatocyte death but also on the inflammatory response subsequent to lipid peroxidation by inhibiting the expression levels of TNF- α mRNAs (Arosio et al., 2000).

2.4 | Antibacterial activity

Staphylococcus aureus is the most common pathogen in cases of bacteraemia, endocarditis, and sepsis. Xiang et al. (2017) showed that aloe-emodin markedly inhibited the biofilm formation, the initial

adhesion, and proliferation stages of *S. aureus* biofilm development in a dose-dependent manner by reducing the production of extracellular proteins. Arylamine NAT activities with paminobenzoic acid and 2-aminofluorene were determined by Wang et al. (1998) in the *Helicobacter pylori* collected from peptic ulcer patients. The results demonstrated that aloe-emodin significantly exhibited the growth inhibition in *H. pylori* cultures in dose-dependent manner. The inhibition reached about 90% when the concentration of aloe-emodin reached 925 μ M. Moreover, there was decreased NAT activity associated with increased aloe-emodin in *H. pylori* cytosols (Wang et al., 1998).

2.5 | Neuroprotective activity

In order to characterize the effects of aloe-emodin as a neuroprotector, an in vitro optic neuropathy model was used by adding N-methyl D-aspartate (NMDA) into N18 retina ganglion cells. The results revealed that aloe-emodin metabolites was able to decrease NMDA-induced apoptosis of retina ganglion cells from 69.9% to 50%. Furthermore, aloe-emodin metabolites significantly decreased the activation of three major MAPK pathways and the activation of downstream genes in nucleus induced by NMDA through regulation of ERK phosphorylation, suggesting that aloe-emodin might be developed as a potential neuroprotector for glaucomatous optic neuropathy (Lin et al., 2007). Tao et al. (2014) was the first to show that aloe-emodin exhibited significant improvement in cognitive deficit in scopolamine-induced amnesia animal model. Besides, it increased the activities of superoxide dismutase, glutathione peroxidase, and the content of acetylcholine whereas decreased the level of MDA and cetylcholinesterase activity. In vitro, aloe-emodin treatment could prevent cytotoxicity in PC12 cells induced by H_2O_2 and reduce significantly the extracellular release of NO, lactate dehydrogenase, and intracellular accumulation of ROS. Therefore, the results indicated that aloe-emodin could have neuroprotective effects against Alzheimer's disease via inhibiting the activity of cetylcholinesterase and modulating oxidative stress (Tao et al., 2014).

2.6 | Antiparasitic activity

Leishmaniasis is a vector-born disease caused by protozoan parasites of the *Leishmania* genus and transmitted by the phlebotomine sandfly (Balasegaram et al., 2012). Dalimi, Delavari, Ghaffarifar, and Sadraei (2015) for the first time demonstrated that aloe-emodin exhibited antileishmanial activity by activation of host macrophages under in vitro and in vivo conditions. Treatment with aloe-emodin significantly inhibited the growth of *Leishmania* amastigotes and induced apoptosis in promastigotes (Dalimi et al., 2015). Moreover, aloe-emodin showed good antiplasmodial activity in a dose-dependent manner. The antiplasmodial EC_{50} value of aloe-emodin was 22 μ g/ml (Kumar, Yadav, Yadav, Rohilla, & Yadav, 2017).

2.7 | Hepatoprotective activity

In a rat model with CCl_4 -induced acute liver injury, aloe-emodin significantly reduced the serum levels of L-aspartate-2-oxoglutarate-

aminotransferase and TNF- α mRNA. The results demonstrated that aloe-emodin was shown to exhibit protective effect not only against hepatocyte death but also on the inflammatory response subsequent to lipid peroxidation (Arosio et al., 2000). In liver fibrosis, hepatic stellate cells are the major cell type responsible for the excessive accumulation of extracellular matrix proteins (Friedman, 1993). Aloe-emodin dose-dependently inhibited Type I collagen production and smooth muscle α -actin expression, which suggest that aloe-emodin may be a useful candidate for developing therapeutic agents for the prevention and treatment of collagen deposition in cases of chronic liver injury (Woo et al., 2002). In another experiment, Lian, Park, Piao, Zhao, and Sohn (2005) utilized rat hepatic stellate cells transformed by simian virus 40 (t-HSC/CI-6) to examine the apoptotic effects induced by aloe-emodin. The findings showed that aloe-emodin inhibited the proliferation of t-HSC/CI-6 cells via the activation of caspase-9 through a mitochondrial pathway (Lian et al., 2005).

2.8 | Other activities

Cárdenas, Quesada, and Medina (2006) found that aloe-emodin suppressed angiogenesis in the chicken chorioallantoic membrane assay in a dose-dependent manner. Moreover, aloe-emodin could inhibit the proliferation of HUVEC in vitro assays. This mechanism might involve the inhibition of the urokinase secretion and tubule formation of endothelial cells (Cárdenas et al., 2006). Treatment with aloe-emodin inhibited adipocyte differentiation and maturation during in vitro human mesenchymal stem cells adipogenesis via the downregulation of adipogenic gene (AP2 and LDL) expression (Subash-Babu & Alshatwi, 2012). The antipsoriatic activity study using Perry's mouse tail model was investigated in vivo by Divya et al. The results indicated that aloe-emodin loaded chitin nanogel could be used for the treatment of psoriasis (Divya et al., 2016). Furthermore, aloe-emodin treatment effectively prevented RIN-5F (pancreatic β -cell) cell from glucotoxicity via regulation of proinflammatory cytokine (IFN- γ , IL-1 β) and downregulation of Bax and caspase-3, suggesting aloe-emodin might be a therapeutic agent to overcome β -cell failure that occurs most on chronic Type 2 diabetic patients (Alshatwi & Subash-Babu, 2016). In a VEGF secretion model, Wu et al. (2016) evaluated the inhibitory effect of aloe-emodin on the hypoxia-induced retinal neovascularization and the possible mechanisms. Aloe-emodin inhibited the VEGFA secretion of ARPE-19 cells under hypoxia condition, decreased the mRNA expressions of VEGFA and PHD-2 in vitro, and prevented hypoxia-induced retinal neovascularization in vivo. The mechanisms underlying this effect might be associated with inhibiting the HIF-1 α /VEGF signaling pathway (Wu et al., 2016). Zang et al. (2017) reported that aloe-emodin-mediated PDT may provide an effective strategy for the treatment of superficial diseases using light sources in the blue region, thus protecting deeper normal tissues from PDT-induced damage (Zang et al., 2017). Furthermore, aloe-emodin could induce the synthesis of matrix proteoglycans, increase the activity of ALP, and enhance the expressions of chondrogenic marker genes such as Collagen II, Collagen X, bone sialoprotein, and runt-related transcription factor 2 via MAPK and bone morphogenetic protein-2

signaling pathways, indicating aloe-emodin may have potential future applications for the treatment of growth disorders (Yang, Li, Heo, & Soh, 2016).

3 | TOXICITY

Vath, Wamer, and Falvey (2002) showed that the exposure of human skin fibroblasts to aloe-emodin and ultraviolet radiation elicited significant phototoxicity. The results indicated a photochemical mechanism involving singlet oxygen to be the most likely pathway responsible for the observed phototoxicity (Vath et al., 2002). In the past 10 years, there have been many reports that showed aloe-emodin exerted hepatotoxicity and nephrotoxicity. It was reported that methanol extracts of *C. occidentalis* seeds exhibited significant cytotoxicities in rat primary hepatocytes and HepG2 cells, indicating that aloe-emodin may be one of its main toxic components (Panigrahi et al., 2015; Panigrahi, Mudiam, et al., 2015). In addition, in vivo studies have showed that aloe-emodin may be the primary chemical component in *R. palmatum* L. that causes hepatic and renal toxicity (Wang et al., 2009). Our group confirmed that aloe-emodin inhibited cell proliferation and induced cell-cycle arrest and apoptosis in HepaRG and HL-7702 cells, most probably through a mechanism involving both Fas death pathway and the mitochondrial pathway by generation of ROS (Dong et al., 2017; Dong, Fu, Yin, Yang, & Ni, 2017). Aloe-emodin treatment could significantly inhibit the liver development of zebrafish and reduce the expression of liver type fatty acid binding protein. The hepatotoxicity mechanism was mainly related to activation of NF- κ B-P53 inflammation-apoptosis pathway but had no effect on the IL-6 and JAK3 (Quan et al., 2019). Moreover, aloe-emodin could induce primary damage to DNA in the liver and kidney of mouse in vivo comet assay (Nesslany, Simar-Meintières, Ficheux, & Marzin, 2009). Zhu et al. (2012) was the first to show that aloe-emodin suppressed HK-2 cell proliferation and induced apoptosis with the involvement of ER stress-triggered signaling pathway. Overall, the implications of the toxic effects of aloe-emodin for clinical use are unclear and these findings underscore the need for risk assessment of human exposure to aloe-emodin.

4 | PHARMACOKINETIC

Lang (1993) were the first to report that oral administration of 14 C-aloe-emodin (4.5 mg/kg) was quickly metabolized to rhein and other unknown metabolites with a terminal half-life in blood of 50 hr then ultimately excreted in urine and feces. Moreover, aloe-emodin rapidly distributed and was eliminated from plasma with a terminal half-life of 78 min after intraperitoneal injection administration at dose of 20 mg/kg, and the drug was still detectable at 4 hr at concentration of 22 ng/ml (Zaffaroni et al., 2003). Yu et al. (2016) investigated the pharmacokinetics of aloe-emodin and relevant metabolites following intravenous and oral administrations to rats. The results showed that the major molecules in blood were aloe-emodin glucuronides and rhein sulfates after intravenous administration. However, when aloe-emodin

was administered orally, aloe-emodin glucuronide and rhein glucuronide were the major molecules, whereas aloe-emodin was not absorbed per se (Yu et al., 2016). It has been shown that aloe-emodin was excreted by first-order kinetics and the concentration-time curve of aloe-emodin was shown to fit two-compartment model after intragastric administration of aloe-emodin (20 and 50 mg/kg) to rats (Feng et al., 2012). In Caco-2 cell model and the everted gut sac model, the absorption rate of aloe-emodin was ranged from 6.60% to 11.32%, and aloe-emodin was mainly absorbed as a glucuronidated or sulfated form, suggesting that a significant amount was transformed during absorption (Park, Kwon, & Sung, 2009).

Fang et al. (2011) suggested that the tissue distributions of aloe-emodin were significantly different between normal and CCl₄-injured rats after successive administration of rhubarb extract for 12 weeks. The concentrations of aloe-emodin in the liver, spleen, and kidney tissues of normal rats were obviously higher than in the model rats (Fang et al., 2011). Zhang et al. (2013) developed a liquid chromatography-mass spectrometry method to detect the pharmacokinetic of aloe-emodin after oral administration of rhubarb peony decoction (RPD). The results showed that the $T_{1/2}$ of aloe-emodin was increased significantly after administration of RPD. In addition, the elimination of aloe-emodin was postponed, indicating that aloe-emodin during absorption and metabolism was possibly an occurred interaction with the compounds in RPD during absorption and metabolism, which resulted in diminishing of toxicity (Zhang et al., 2013). In another experiment, the concentrations of aloe-emodin in rat serum were much lower than those of chrysophanol after oral administration of Da-Cheng-Qi decoction, indicating that aloe-emodin has a high binding activity to organs and a low absorption and blood distribution (Tang et al., 2010).

5 | FUTURE PERSPECTIVES AND CONCLUSIONS

Aloe-emodin is an anthraquinone with potential pharmacological benefits, including anticancer, anti-inflammatory, antiviral, antibacterial, antiallergic, antiosteoporotic, antidiabetic, immunosuppressive, neuro-protective, and hepatoprotective effects. It is suggested that aloe-emodin may be a valuable therapy option for the prophylaxis and treatment of various diseases, including influenza virus, inflammation, sepsis, Alzheimer's disease, glaucoma, malaria, liver fibrosis, psoriasis, Type 2 diabetes, and growth disorders. Moreover, previous studies have exhibited that aloe-emodin in conjunction with various chemotherapeutic agents and targeted therapies are more effective in the treatment of cancer. Hence, it is important to evaluate the effect of aloe-emodin on other drug metabolism before aloe-emodin can be safely used in combination with other drugs.

Aloe-emodin strongly suppresses the expression of proteins and genes, such as the casein kinase II, ALP, c-Myc, calpain-2, ubiquitin-protein ligase E3A, ER α , NAT, NF- κ B, ataxia telangiectasia-mutated gene, ataxia telangiectasia-mutated and Rad3-related, breast cancer gene 1, O⁶-methylguanine-DNA-methyltransferase, and FAK. However, the expression levels of Bax, Fas, p53, caspase-3, ERK1/2,

TABLE 1 Pharmacology of aloe-emodin

Pharmacological effect	Cell lines/model	Activity/mechanism(s) of action	Application	Reference
Anticancer activity	CH27 cells	Bax and Fas death pathways	In vitro	Lee et al. (2001)
	H460 cells	Caspase-3-dependent pathway	In vitro	Lai et al. (2007) Lee et al. (2005)
	SCC cells	MAPKs and PI3K/AKT pathways	In vitro	Yeh et al. (2003)
	AGS and NCI-N87 cells	Activating the expression of caspase-3 and suppressing the activity of the casein kinase II	In vitro	Wu et al. (2017)
	MGC-803 cells	Decreases ALP activity	In vitro	Chen et al. (2007)
	SGC-7901 cells	PKC pathway and c-Myc suppression; Induces apoptosis and necrosis; The mitochondrial pathway	In vitro	Guo, Xiao, Liu, et al. (2007)
	MKN45 cells	Induces cell cycle arrest at G0/G1 phase	In vitro	Guo et al. (2008); Li et al. (2016); Lin et al. (2017)
	HepG2 cells	Increases the expression of Fas/APO1 receptor and Bax; Enhances the intracellular level of reactive oxygen species and oxidation of PRDX-2 and -4; JNK activation	In vitro	Chihara et al. (2015)
	Hep3B cells	Induces p21 expression; Decreases the stability of CARP mRNAs	In vitro	Kuo et al. (2002); Lu, Shen, Chung, and Ong (2007); Lu, Shen, Ong, and Chung (2007)
	Huh-7 cells	Downregulates CAPN2 and UBE3A proteins	In vitro	Kuo et al. (2002); Lin et al. (2011)
	MCF-7 cells	Inhibits cell proliferation, adhesion, invasion, and induces apoptosis; Downregulates ER α protein levels; Ras/ERK and PI3K/mTOR pathways	In vitro	Jeon et al. (2012)
	HER-2 cells	Akt/mTOR signaling pathway	In vitro	Chen et al. (2016); Huang et al. (2013); Tseng et al. (2017)
	A375.S2 cells	Inhibits the mRNA expression and enzyme activity of NAT	In vitro	Ma et al. (2016)
	B16 cells	Activates ERK1/2	In vitro	Lin et al. (2005)
	A375 cells	Inhibits ERK	In vitro	Radovic et al. (2012)
	B16-F10 cells	Reduces of MMP-9 activity	In vitro	Radovic et al. (2012)
	SK-MEL-110 and SK-MEL-28 cells	Increases the expression of inflammation-associated factors	In vitro	Tabolacci et al. (2010)
	HL-60 cells	activation of caspase-3	In vitro	Tabolacci et al. (2015)
	U937 cells	Enhances MMP secretion	In vitro	Chen et al. (2004)
	WiDr cells	Inhibits the nuclear translocation and DNA binding of NF- κ B	In vitro	Tabolacci et al. (2011)
	SW 480 and SW 620 cells	Downregulates the phosphorylation of ERK1/2 and activates SAPK/JNK and p38MAPK	In vitro	Suboj, Babykutty, Gopi, et al. (2012)
	Apc-deficient Min mouse model	Reduces cell proliferation	In vivo	Suboj, Babykutty, Srinivas, and Gopala (2012)
	HeLa cells	Induces G2/M and S phase cell cycle arrest and improves ALP activity; PKC α pathway	In vitro	Shimpo et al. (2014)
	SCC-4 cells	Inhibits the gene expressions of ATM, ATR, BRCA1, and MGMT; The Fas/death-receptor and mitochondria pathway	In vitro	Luo et al. (2014); Guo, Xiao, Zhang, et al. (2007)
	U-373MG cells	PKC signaling pathway	In vitro	Chen et al. (2010); Chiu et al. (2009)
	C6 glioma cells	Inhibits the activation of ERK1/2	In vitro	Acevedo et al. (2004)
	U87 cells	The mitochondria pathway	In vitro	Mijatovic et al. (2005)
	HO-8910PM cells	Reduces the protein and mRNA expression of FAK	In vitro	Ismail et al. (2013)
	T24 bladder cells	Activation of p53, caspase-3	In vitro	He et al. (2008)
	NPC cells	The mitochondria pathway	In vitro	Lin et al. (2006)
	PC3 cells	Targets mTORC2	In vitro	Lin et al. (2010)
	TE1 cells	Inhibits AKT and ERK	In vitro	Liu et al. (2012)
	MG-63 cells	ROS-JNK signaling pathway	In vitro	Chang et al. (2016)
Antivirus activity	Influenza A	Upregulated galectin-3 and thioredoxin	In vitro	Tu et al. (2016)
	Japanese encephalitis virus and enterovirus 71	Induces the expression of IFN- α , PKR, and OAS	In vitro	Li et al. (2014)
				Lin et al. (2008)

(Continues)

TABLE 1 (Continued)

Pharmacological effect	Cell lines/model	Activity/mechanism(s) of action	Application	Reference
Anti-inflammatory Activity	LPS-stimulated RAW 264.7 cells	Inhibits NF- κ B, MAPK, and PI3K pathways	In vitro	Hu et al. (2014)
	CCl ₄ -induced acute liver injury rats	Inhibits the expression levels of TNF- α mRNAs	In vivo	Arosio et al. (2000)
Antibacterial activity	<i>Staphylococcus aureus</i>	Reduces the production of extracellular proteins	In vitro	Xiang et al. (2017)
	<i>Helicobacter pylori</i>	Decreases NAT activity	In vitro	Wang et al. (1998)
Neuroprotective activity	Retina ganglion cells	Regulates ERK phosphorylation	In vitro	Lin et al. (2007)
	Scopolamine-induced amnesia animal model	Increases the activities of SOD, GPx, and the content of ACh, decreases the level of MDA and AChE activities	In vivo	Tao et al. (2014)
	H ₂ O ₂ -induced PC12 cells	Inhibits the activity of AChE and modulates oxidative stress	In vitro	Tao et al. (2014)
Antiparasitic activity	Leishmania	Activation of host macrophages	In vitro and vivo	Dalimi et al. (2015)
Hepatoprotective activity	CCl ₄ -induced acute liver injury rats	Reduces the serum levels of L-aspartate-2-oxoglutarate-aminotransferase and TNF- α mRNA	In vivo	Arosio et al. (2000)
	Hepatic stellate cells	Inhibits type I collagen production and smooth muscle α -actin expression	In vitro	Friedman (1993)
	t-HSC/CI-6	The mitochondrial pathway	In vitro	Lian et al. (2005)

TABLE 2 Toxicity of aloe-emodin

Cell lines/model	Activity/mechanism(s) of action	Application	Reference
Human skin fibroblasts	Singlet oxygen	In vitro	Vath et al. (2002)
HepG2 cells	Inhibits the proliferation	In vitro	Panigrahi, Mudiam, et al. (2015)
HepaRG cells	Fas death pathway and the mitochondrial pathway	In vitro	Dong, Fu, Yin, Qu, et al. (2017)
HL-7702 cells	Fas death pathway and the mitochondrial pathway	In vitro	Dong, Fu, Yin, Yang, and Ni (2017)
The liver and kidney of mouse	Induces primary damage to DNA	in vivo	Nesslany et al. (2009)
HK-2 cell	ER stress-triggered signaling pathway	In vitro	Zhu et al. (2012)

MMP, IL-2, IL-12, GM-CSF, and IFN- γ are upregulated. Moreover, aloe-emodin affects the MAPKs, PKC, NF- κ B, Ras/ERK, ROS-JNK, PI3K/AKT/mTOR, and mitochondrial signaling pathways. The evidence provided strongly supports the ability of aloe-emodin to inhibit diverse events involved in cell apoptosis, cell metastasis, and cell-cycle arrest and clearly supports its traditional use in the treatment of cancers. This paper summarizes the pharmacological properties and molecular mechanisms of aloe-emodin (Table 1) and strongly supports the view that emodin has a broad application prospect.

In addition, there have been adverse effects reported on aloe-emodin, including the phototoxicity, hepatotoxicity, and nephrotoxicity (Table 2). Aloe-emodin shows hepatotoxicity and nephrotoxicity by inducing apoptosis in the Fas death, mitochondrial, and ER stress-triggered signaling pathway. The compatibility of traditional Chinese medicine can reduce the toxicity of aloe-emodin. However, its mechanism needs to be further revealed. Therefore, further studies are needed to attenuate the toxicity of aloe-emodin.

The poor intestinal absorption, short elimination half-life, and low bioavailability in vivo of aloe-emodin limit its clinical application further. Traditional oral administration would not be the best method of

application of aloe-emodin. The pharmacokinetics of aloe-emodin could be improved by inhibiting its glucuronidation metabolism. However, we should pay attention to the ability of aloe-emodin to lead to other toxicity problems due to the increase in its content in further research.

This paper potently supports the view that aloe-emodin has a broad application prospect. Nevertheless, further researches are needed to improve oral bioavailability, attenuate the side-effects, and enhance its efficiency.

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CONFLICT OF INTEREST

The authors have declared that there is no conflict of interest.

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